Tut-12

1b. vertex cover $\leq_p$ Independent set
Independent $\leq_p$ clique
clique $\leq_p$ vertex cover
3-SAT $\leq_p$ clique
3-SAT $\leq_p$ vertex cover
3-SAT $\leq_p$ Independent set

HC $\leq_p$ HP
HP $\leq_p$ TSP
vertex cover $\leq_p$ SC
SC $\leq_p$ hitting set
subset sum $\leq_p$ knapsack
partition $\leq_p$ subset sum

2) Heuristic: construct a DFS spanning tree of G, return all non-leaf vertices

a) TO prove: Heuristic returns a vertex cover

Proof: consider any edge $(u,v) \in E$

case 1: Tree edge
one vertex is parent other is child
If child is a leaf → parent is non-leaf → includes
If child is non-leaf → already included.

case 2: Back edge
connects a node to its ancestor
Ancestor has children → non-leaf → includes

Every edge has at least one endpoint in the set → valid vertex cover.

b) let
S = set of non-leaf nodes
each non-leaf node has atleast one child
define edges from node to one child
These edges are disjoint (no shared endpoints)
Any vertex cover must include ≥ 1 endpoint per edge

so

$$opt \geq \frac{|s|}{2} \Rightarrow |s| \leq 2 \cdot opt$$

c) consider path graph with even no of vertices

$$v_1 - v_2 - v_3 - v_4$$

Heuristic selects: $v_2, v_3 \rightarrow$ size = 2

optimal: $v_2 \rightarrow$ size $=$)

$$\frac{2}{1} = 2$$

4) 1) Normalize speeds

slow machine speed = 1

fast machine speed = 2

2) Greedy list scheduling

sort jobs in decreasing t_i

Assign each job to machine with minim

current load

correctness:

let:

opt = optimal makespan

T = algorithm makespan

from scheduling theory

$$T \leq \frac{\sum t_i}{m+2k} + \max t_i$$

$$opt \geq \max \left(\frac{\sum t_i}{m+2k}, \max t_i \right)$$

$$T \leq 3 \cdot opt$$

$T(n) = O(n \log(m+k))$